WasteSort: from dataset to prediction

Goal Diggers

Saturday demonstration
S-CSIS311 · BCS34

Six material classes
One uploaded photo
Measured results and limitations



Prepare the data before training

Partition	Images	Role
Training	1,774	Fit parameters
Validation	371	Choose settings and model
Test	376	Evaluate the frozen choice

Six excluded files · 45 reviewed candidate pairs · seed 42

Compare four primary approaches

Approach	Representation
SVM and Random Forest	1,764 HOG features
Small CNN	160 × 160 RGB images
MobileNetV2	Pretrained image features

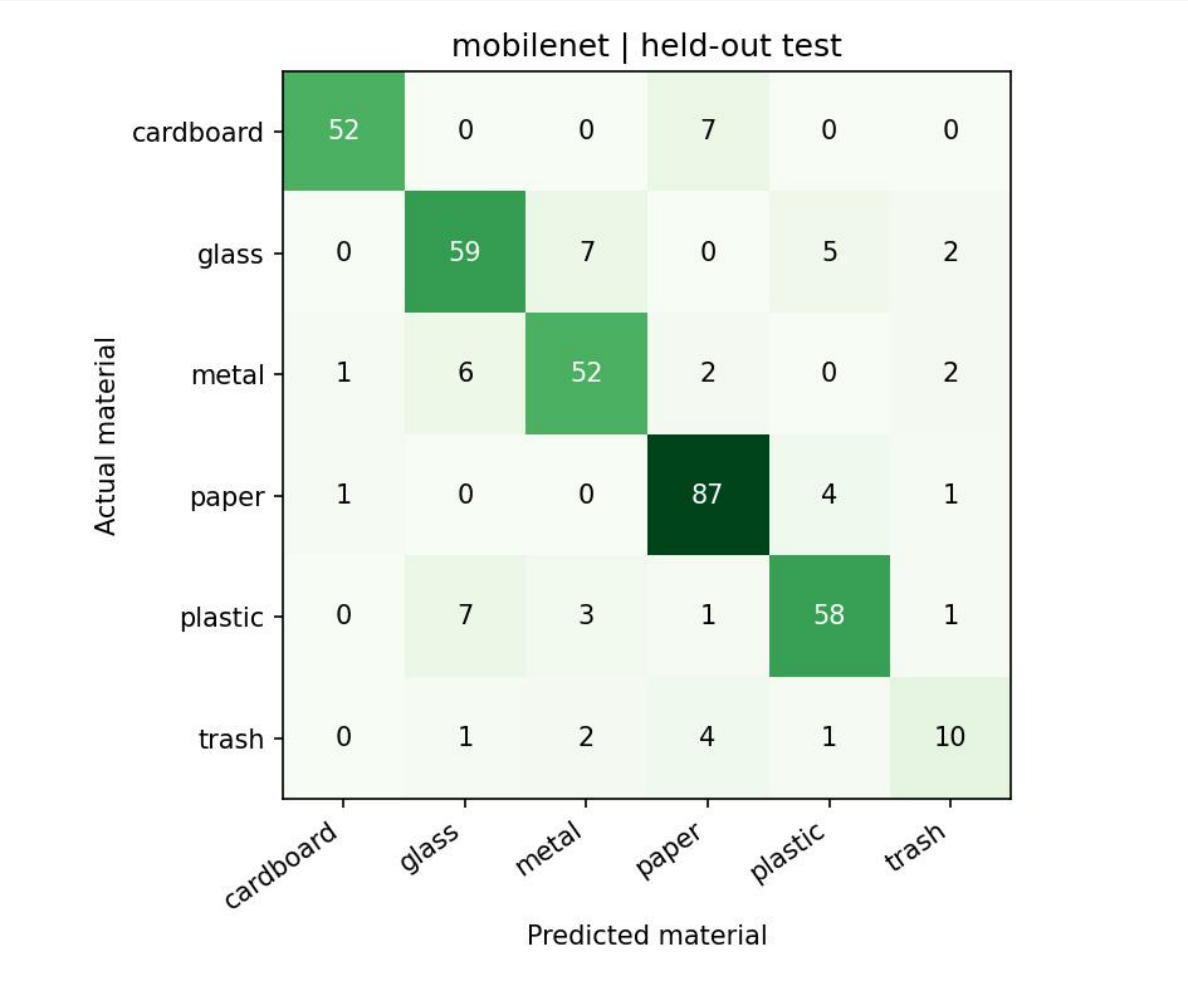
Exercises: KNN, Decision Tree and separate Linear Regression

MobileNetV2 led the published comparison

Model	Validation F1	Test accuracy	Test F1
HOG + SVM	60.62%	65.96%	63.52%
HOG + Random Forest	55.94%	63.56%	60.22%
Small CNN	62.07%	59.57%	57.68%
MobileNetV2	84.74%	84.57%	81.22%

Macro F1 shown · one split and seed · weighted CNN is a separate ablation

Overall accuracy hides missed trash items



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trash images recognized
55.56% recall

Check errors by class,
not just total accuracy.

Demonstrate the app and explain its limits

1. Start the app
2. Upload one clear item photo
3. Select Identify material
4. Explain and download the result

Before choosing a bin

Check posted campus rules.

No campus-photo accuracy claim.

Unrelated objects can receive a waste label.

AI-assisted group work · methodology changes documented for instructor review